Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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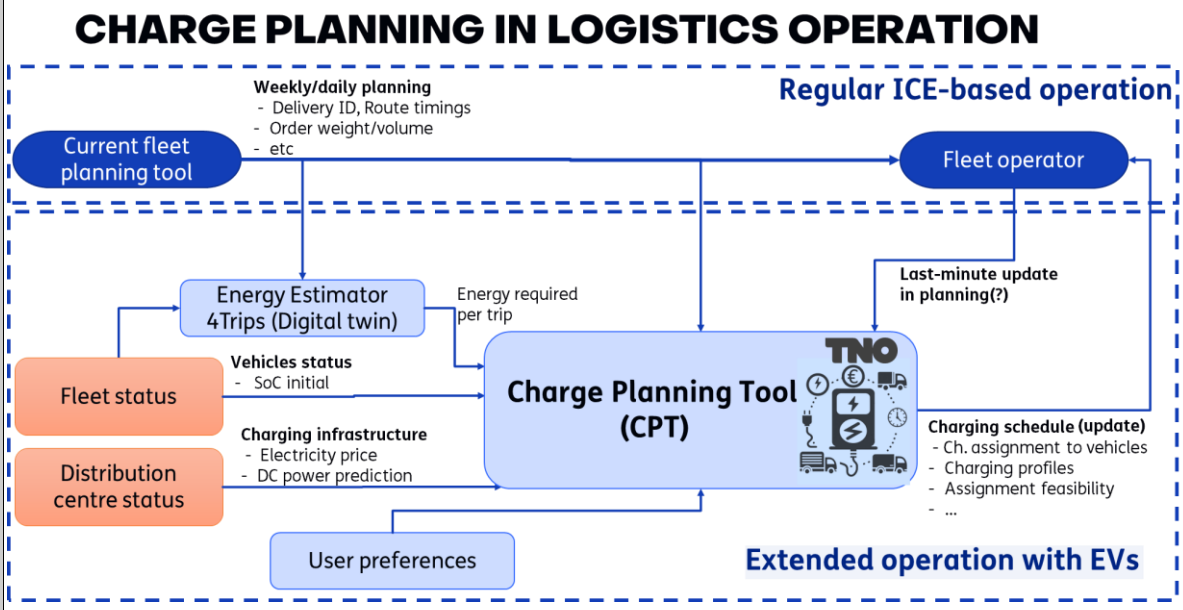
2. Technology Overview

Title of Proposed Technology	Battery-aware EV Charge Planning and Digital Twin Platform for Smart Charging and Grid-constrained Fleets			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Artificial Intelligence ( ), Advanced Bio ( ), Carbon Neutrality ( ), Hydrogen ( ), Cybersecurity ( ), Space and Aerospace ( ), Quantum ( ), Advanced Robotics and Manufacturing ( ), Next-generation Nuclear Power ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW	Heavy Electrical Equipment	Power conversion equipment
Concept of Proposed Technology	**Background and motivation** For scalable deployment across multiple charger vendors and vehicle models, interoperability standards are essential. OCPP enables vendor-neutral communication between charge points and central systems [3]. ISO 15118 supports smart charging and Plug and Charge between the EV and the charging station [4]. Many depots also evaluate stationary Battery Storage Systems (BSS) to buffer charging power			

peaks and reduce grid connection and demand charges. TNO publications describe fleet charging optimization that explicitly exploits a BSS while accounting for battery degradation and grid constraints [5].



Accurate prediction of charging time and power demand requires battery-aware models that capture aging and temperature effects. Rath and Wilkins (2024) report an adaptive battery digital twin method for EV charge profiles designed to improve fleet charge scheduling [1].

Fleet electrification is increasingly constrained by limited grid connection capacity and the need to coordinate many vehicles charging in the same time window. TNO's Charge Planning Tool addresses this by jointly optimizing logistics planning and charging while considering grid congestion [6].

Proposed solution

TNO proposes to co-develop a production-oriented software platform that couples:

- A battery and vehicle digital twin (DT) that predicts state of charge, charge time, and operational energy use for each vehicle and charging session.

- A fleet and site level charge planning optimiser that computes charger assignment, charging power profiles, and timing under grid, tariff, and operational constraints.

Digital twin core

The DT will combine physics-based battery models (electrical and thermal behaviour) with data assimilation from real charging data, enabling fast prediction with low training cost. TNO has

demonstrated battery-DT-based charge profile prediction with <2% charge time error and a 7% improvement versus a standard charge profile in a real case study [1].

Charge planning core

The optimiser will generate charging schedules for fleets with heterogeneous arrival and departure times, limited charging infrastructure, and grid connection constraints. A TNO algorithm (URBANIZED case study) achieved 13% savings in total charging cost and runtime <160 seconds for 200 vehicles, enabling real-time operations [2].

System integration scope

The joint platform is designed to integrate with common charging management system architectures via Open Charge Point Protocol (OCPP) interfaces and to support vehicle to charger authentication via ISO 15118 Plug and Charge where available [3,4].

Coverage of EV infrastructure and stationary battery systems

In addition to direct EV charging, the optimiser will include optional coordination with on-site stationary battery energy storage systems (BSS) to mitigate grid congestion and reduce operational cost. TNO has published optimisation approaches for EV fleets that explicitly consider Battery Storage Systems and grid-related constraints [5].

Expanded joint development scope (technical boundary conditions)

In-scope (jointly developed software components and interfaces)

- Charging infrastructure interoperability through OCPP (incl. smart charging capabilities) for vendor-neutral telemetry and control, with optional ISO 15118 Plug and Charge workflows where supported [3,4,11].

- Data ingestion and harmonisation for charging sessions, charger status, metering, site grid constraints, dynamic tariffs, and vehicle operational metadata to enable consistent optimisation across sites [6,11].

- Battery and vehicle digital twin layer for fast prediction of charging power and time, including online parameter updates from charging session data and degradation-aware constraints [1,13].

- Fleet and site charge planning and real-time re-planning that co-optimises charging feasibility,

	energy cost, and battery lifetime under grid congestion, using optimisation approaches demonstrated in Wilkins co-authored work [2,6]. - Optional co-optimisation with on-site stationary storage (BSS) and renewables to shave peaks and shift energy, aligned with prior work on BSS-supported charging and fast-charger plus BSS architectures [5,12]. - Operator-oriented reporting outputs for integration into the partner's EV performance reporting product line, including plan feasibility, missed-departure risk, peak-demand drivers, energy cost drivers, and degradation-related KPIs (methodology defined in WP1) [2,6]. Out-of-scope (not part of this joint development) - Hardware redesign of chargers, vehicles, or batteries; consumer-facing payment and billing platforms; and embedded BMS firmware development (except for defined data interfaces).
Rationale for Joint R&D Project	Why joint R&D is needed **Partner context:** The SME partner is developing smart charging and load management software and is exploring fleet charge planning and vehicle digital twin functions, as mapped in the TIPA Korean SME scoping dataset [11]. Fast EV adoption is constrained by limited grid connection capacity and high operational complexity at charging sites. TNO's Charge Planning Tool shows how digital twins of vehicles, combined with measured energy use and route information, can predict arrival timing and state of charge for each vehicle and enable robust fleet charging plans under grid limits [6]. The SME partner operates or supports EV charging operations and is developing advanced smart charging functions, including load management, charge planning, and fleet performance reporting. The partner requires validated battery-aware prediction and optimisation methods to increase schedule robustness, reduce peak power, and support scalable deployment across many sites. TNO contributes validated models, optimisation methods, and testing infrastructure. The collaboration will translate scientific results into an integrated prototype and pilot validation using real operational data from the SME partner. This directly supports international joint R&D objectives by combining applied research with commercial deployment capability.
Key Words	1. Smart charging 2. Charge planning

	3. Battery digital twin 4. Grid congestion mitigation 5. Fleet energy optimisation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	R&D objectives - Develop a battery and vehicle DT component for charge time and state-of-charge prediction, including online parameter updating from charging session data [1]. - Develop a site and fleet charge planning optimiser that is battery-degradation-aware and grid-constrained, supporting real-time re-planning [2]. - Integrate with charging management systems (CSMS) using OCPP and optional ISO 15118 Plug and Charge workflows [3,4]. - Define a reference data model and API specification for integration with charging management systems and analytics dashboards, including cybersecurity and data governance requirements (agreed in WP1) [3,11]. - Extend the optimiser scope to include optional co-optimisation with on-site stationary storage and renewable generation where available, leveraging published approaches for BSS-assisted charging and peak mitigation [5,12]. - Develop an evaluation and verification methodology for uncertainty-aware planning (arrival SOC uncertainty, charger availability, tariff volatility) and for safety margins that protect operational robustness and battery lifetime [1,2,6]. - Validate the combined DT and optimiser in (i) simulation with historical session data, (ii) hardware-in-the-loop and climate-controlled testing, and (iii) a pilot using the SME partner's operational environment.			
R&D Target Specifications	Target specifications (to be verified in pilot) - Charge time prediction mean absolute percentage error (MAPE) <= 5% on representative EV charging sessions. Reference performance: <2% error demonstrated using a battery DT in a case study [1].			

	- Fleet scheduling runtime <= 5 minutes for 200 vehicles and multiple chargers; reference performance: <160 seconds for 200 vehicles in a published schedule algorithm [2]. - Peak grid power reduction >= 10% for constrained sites by load shifting, charger power modulation, and optional BSS coordination [5,6]. - Reliability: >= 99% of planned departures meet required state-of-charge thresholds in simulation and pilot (definition agreed in WP1). - Interoperability: prototype demonstrates end-to-end control loop with at least one CSMS via OCPP smart charging functions, and optional ISO 15118 Plug and Charge handshake in a lab environment (where supported) [3,4]. - Re-planning latency: ability to re-optimise after disruptions (delay, unexpected consumption, charger outage) within operationally relevant time, consistent with published real-time scheduling performance for 200-vehicle fleets [2,6].
R&D Plan	R&D plan (36 months, indicative) **WP1 Requirements, data, and system architecture (M1-M4)** - Define use cases, KPIs, and constraints for fleet, depot, and public charging. - Specify data interfaces (charging sessions, charger status, tariffs, grid limits, vehicle metadata) and integration with CSMS. - Define pilot dataset structure, anonymisation approach, and a baseline planning workflow for comparison (business-as-usual). - Define scope boundaries, in-scope and out-of-scope items, and a joint verification plan with test cases for interoperability, robustness, and cybersecurity. **WP2 Digital twin development and calibration (M3-M16)** - Implement battery and vehicle DT models for fast charge profile prediction. - Calibrate and validate DT using partner charging data; implement continuous updating. **WP3 Charge planning optimisation (M8-M24)** - Implement schedule optimisation with battery degradation cost terms and grid constraints. - Add optional coordination with on-site BSS and renewables. **WP4 Integration and test (M22-M30)** - Integrate DT and optimiser into a deployable software stack (API-driven microservices). - Validate using TNO testing infrastructure, including climate-controlled battery and pack testing

	with high-power cycling capabilities [7,8]. - Perform software-in-the-loop and hardware-in-the-loop style integration tests with a representative charger interface and synthetic fleet scenarios before pilot deployment [11]. - Implement deployment packaging (containers) and monitoring hooks to support repeatable testing and pilot operation, including logging of optimisation decisions for traceability. **WP5 Pilot demonstration and evaluation (M28-M36)** - Deploy prototype in a pilot environment with the SME partner and evaluate KPIs. - Run an A/B evaluation against a baseline planning approach to quantify benefits and limitations (KPIs agreed in WP1), and capture operational feedback for productisation [6]. - Produce final technical report and exploitation plan.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO expected roles and responsibilities - Technical lead for digital twin modelling and charge planning optimisation. - Provide validated algorithms and method implementation based on published work on charge profile prediction [1] and real-time fleet scheduling with battery degradation considerations [2]. - Provide test and validation capability, including Battery Lab infrastructure and controlled test environments for batteries, packs, and EV-level testing [7,8]. - Lead evaluation methodology, KPI definition, and dissemination (joint publications, workshops).
Expected R&R of Korean SME Partner	SME partner expected roles and responsibilities - Provide operational requirements, user workflows, and constraints from charging operations. - Provide relevant datasets (charging sessions, charger telemetry, tariffs, grid capacity constraints) and support secure data access. - Provide integration support with the partner's charging management platform (CSMS), including OCPP message flows and deployment constraints [3]. - Host and support pilot demonstration, including monitoring, feedback, and operational acceptance testing.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes - Integrated prototype linking battery and vehicle digital twins with charge planning optimisation for EV infrastructure and optional BSS coordination. - Verified performance against agreed KPIs on both historical data and pilot deployment. - Technical documentation and integration guidelines for CSMS deployment using OCPP and optional ISO 15118 Plug and Charge flows [3,4]. Expected impact - Improved utilisation of constrained grid connections by scheduling and power modulation. TNO has shown that charging plans can be generated for hundreds of vehicles within minutes while considering grid connection limits and digital twin based energy prediction [6]. - Reduced total cost of charging and improved battery lifetime by integrating battery degradation cost terms in charging schedules [2]. - Stronger competitiveness of the SME partner through differentiated smart charging functionality and scalable deployment. - Contribution to sustainable mobility and grid congestion mitigation via optimised charging operations [6].

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization